

United International University

IPE 3401: Industrial and Operational Management

Assignment 2 (Spring 2026 Trimester)

Problem 1

Suppose Mr. Jon Snow wants to make 5 type of knives from a material called dragon glass. First you need to cut the glasses in shapes and then you need to do some drilling on them. But he doesn't know how to sequence his job and you are a friend of him, who has studied IPE 401 course and know how to apply Johnson's rules for sequencing the jobs. Now sequence the jobs and draw the "**Gant Chart**" and calculate the idle time.

Job	Processing Time for Jobs (Days)	
	Work Station 1 (Cutting)	Work Station 2 (drilling)
A	5	3
B	8	2
C	4	1
D	7	8
E	12	6

Problem 2

Sequence the following jobs of a tailor who needs a cutting machine and a sewing machine to make dresses. He needs to cut the cloths first and then it can be sewed. Apply Johnson's rule to sequence the jobs, draw a "Gantt chart" and find the idle time for both machines.

Job	Processing Time of Work center 1	Processing time of Work center 2
C	7	6
A	6	4
F	10	7
D	2	3
E	5	1
B	6	5

Problem 3

For the following order table, select the best sequence of jobs to be processed based on SPT and EDD rules, with logic behind your selection:

Job	Processing time (day)	Due Date (day)
A	3	4
B	5	7
C	4	8
D	6	9
E	2	5

Problem 4

During the past 8 years the port of Baltimore has unloaded large quantities of grain. ($\alpha = 0.3$) The forecast of 2012 was 369 unit. Find the forecast for the 2017 and 2011.

Year	2009	2010	2011	2012	2013	2014	2015	2016
Actual	395	359	379	389	398	378	387	396

Problem 5 (Least squares/Linear Regression)

The demand for electrical power at EGCB over the year 2007-2013 is given. Find the **overall trend**. Also find the demand of power at year 2101.

Year	2007	2008	2009	2010	2011	2012	2013
Demand	1100	1115	1190	1200	1240	1170	1315

Problem 6

You have got a full-time job in sales in a manufacturing company. The manager gave you smoothing constant = 0.35 and forecast value of 2015 which was 218. Find MSE of the following years.

Year	2015	2016	2017	2018	2019	2020	2021
Actual	245	239	242	267	221	275	270

Problem 7

A company records the following monthly demand for a product:

Month	1	2	3	4	5
Demand	215	235	224	218	225

Find the following:

- i) The forecast for Month 2 through Month 6, using the exponential smoothing method with a smoothing constant of 0.4.
- ii) The Mean Absolute Deviation (MAD) and Mean Squared Error (MSE) for Months 2–5.

Problem 8

Sunrise Furniture Ltd. manufactures wooden chairs. The company has provided the following required production units for the upcoming year:

Quarter	Q1	Q2	Q3	Q4
Required Production	3,800	5,400	6,200	4,600

Some additional information about the direct material is as follows:

- Each unit requires 5 kg of wood
- Wood purchase price = Tk. 55/kg
- Desired ending direct materials (DM) inventory = 15% of next quarter's production need
- Beginning DM inventory (Q1) = 2,500 kg; Desired ending DM inventory (Q4) = 3,000 kg

Prepare the Direct Materials Budget for all four quarters, showing:

- i) DM to be purchased (in kg), and
- ii) Total DM purchase cost.

Problem 9

BrightGlow Ltd. produces scented candles. Sales forecast is as follows:

Quarter	Sales Units
Q1	4,500
Q2	7,200
Q3	8,500
Q4	5,800

- Desired ending FG = 12% of next quarter's sales
- Beginning FG (Q1) = 480 units; Ending FG (Q4) = 550 units
- Each unit requires 2.5 labor hours at Tk. 80/hour

Required: Prepare the Direct Labor Budget.

Problem 10

NovaStar Plastics Ltd. manufactures plastic water bottles. The company has provided the following required production units for the upcoming year:

Quarter	Q1	Q2	Q3	Q4
Required Production	8,500	11,200	13,600	9,400

Some additional information about the direct material is as follows:

- Each unit requires 4 kg of plastic resin
- Plastic resin purchase price = Tk. 35/kg
- Desired ending direct materials (DM) inventory = 12% of next quarter's production need
- Beginning DM inventory (Q1) = 3,800 kg; Desired ending DM inventory (Q4) = 4,200 kg

Prepare the Direct Materials Budget for all four quarters, showing:

- i) DM to be purchased (in kg), and
- ii) Total DM purchase cost.

Problem 11 (ABC)***

Stellar Shoes Ltd. produces two products — Economy Shoes and Premium Shoes — with the following data:

Item	Economy Shoes	Premium Shoes
Annual production	2,000 units	800 units
Direct material/unit	Tk. 350	Tk. 650
Direct labor/hr	Tk. 250	Tk. 450
DL hours/unit	3 hrs	3 hrs

Overhead activities:

Activity	Total Cost (Tk.)	Cost Driver	Economy	Premium	Total
Machine setups	72,000	Setups	30	90	120
Quality control	90,000	Inspection hrs	150	350	500
Packaging	54,000	Packing orders	60	140	200

Required: (a) Unit cost under Traditional Costing. (b) Activity rates and unit cost under ABC. (c) Which product is over/under-costed?

Problem 12 (ABC)***

GreenLeaf Foods Ltd. produces two products — Regular Biscuits and Deluxe Biscuits — with the following data:

Item	Regular Biscuits	Deluxe Biscuits
Annual production	3,000 units	600 units
Direct material/unit	Tk. 280	Tk. 520
Direct labor/hr	Tk. 200	Tk. 400
DL hours/unit	2 hrs	2 hrs

Overhead activities:

Activity	Total Cost (Tk.)	Cost Driver	Regular	Deluxe	Total
Machine setups	48,000	Setups	20	80	100
Inspection	75,000	Inspection hrs	300	450	750
Material handling	36,000	Moves	90	110	200

Required: (a) Unit cost under Traditional Costing. (b) Activity rates and unit cost under ABC. (c) Which product is over/under-costed?

Problem 13

A manufacturing firm produces Component X and Component Y. Each unit of Component X requires 4 hours of processing and 2 hours of assembly. Component Y requires 8 hours of processing and 1 hour of assembly. The workshop has a maximum weekly capacity of 40 processing hours and 8 assembly hours. Due to a contract, they must produce at least 2 units of Component X. Additionally, the total units produced must be at least 3. If Component X earns \$200 and Component Y earns \$350, how many of each should be made to maximize profit? Solve the problem using graphical techniques. (No graph paper will be provided.)

Problem 14

Two friends, A and B, decide to play an Even–Odd finger game. Each simultaneously shows either an even number of fingers or an odd number of fingers. The rules are: if both show even numbers, A wins 3 points; if both show odd numbers, B loses 2 points; if A shows even and B

shows odd, B wins 1 point; and if A shows odd and B shows even, A loses 1 point. The game is zero-sum, meaning any points gained by A are lost by B. Determine whether there is a saddle point, and if not, calculate the optimal mixed strategies for both players and the expected value of the game.

Problem 15

Consider a game between two players, P1 and P2. Each player simultaneously chooses one of three strategies: P1 can choose A, B, or C, and P2 can choose X, Y, or Z. The gains or losses depend on the combination of strategies chosen. If P1 plays A and P2 plays X, P1 gains \$4; if P1 plays A and P2 plays Y, P2 gains \$2; if P1 plays A and P2 plays Z, P1 loses \$1. If P1 plays B and P2 plays X, P2 loses \$3; if P1 plays B and P2 plays Y, P1 gains \$5; if P1 plays B and P2 plays Z, P1 gains \$3. If P1 plays C and P2 plays X, P1 loses \$2; if P1 plays C and P2 plays Y, P2 gains \$4; and if P1 plays C and P2 plays Z, P1 gains \$3. Formulate a payoff table and find the optimal strategy.

Problem 16

Rafi and Sami play a Card-Pick game in which each player simultaneously chooses either Red or Black. The rules are: if both choose Red, Rafi wins \$5; if both choose Black, Rafi wins \$2; if Rafi chooses Red and Sami chooses Black, Sami wins \$3; and if Rafi chooses Black and Sami chooses Red, Sami wins \$4. The game is zero-sum, meaning that any points Rafi gains are points Sami loses. Determine whether there is a saddle point, and if not, calculate the optimal mixed strategies for both players and the game's expected value.

Problem 17

Nina and Mila play a Number-Call game in which each player simultaneously chooses either Odd or Even. The rules are: if both choose Odd, Nina wins 6 points; if both choose Even, Mila wins 4 points; if Nina chooses Odd and Mila chooses Even, Mila wins 2 points; and if Nina chooses Even and Mila chooses Odd, Nina wins 3 points. The game is zero-sum, meaning that any points Nina gains are points Mila loses. Determine whether there is a saddle point, and if not, calculate the optimal mixed strategies for both players and the game's expected value.

Problem 18

Find the optimal strategy that both players should play?

Strategy	P2-1	P2-2	P2-3
P1-1	-3	4	2
P1-2	1	-2	5
P1-3	4	3	-1